

toolbar on the right side of the main window allows you to select from amongst a large number of predefined components like transmission lines, wires, resistors, capacitors, inductors, semiconductor devices (diode, BJT, JFET and MOSFET), transistors, op-amps, voltage and current source, test points, ground, transformers, etc. You can even add readymade logic gates, comparators, latches, flip-flops, counters and even include user-defined parameters. But the option to save digital logic circuits is available only in registered version of the software.

Click on the desired components and click where on the drawing board you would like to include them. You can add multiple copies of the same component this way. You can deselect the component by right clicking anywhere on the workspace, and perform necessary editing to the added component or include a new type of component in the design. Right click on the component to move them, edit the parameters and rotate or flip as required. You can add text notes relating to each circuit by clicking on the *NOTES* tab. These are saved along with the file. Once the circuit is completely built, the analysis can be performed.

As mentioned in the features, this software has a wizard that allows the user to check whether the design entered matches the default designs given in the library of this program. This option comes in the menu bar under *Analyse > Project Defaults Wizard*.

If you are not ready to create your own circuit, a set of demo files are already included in the library of the program, to assist the users. These can be opened by going to *File > Open Demo* and selecting the required file. Fig. 2 shows such a loaded demo schematic of astable circuit, the transient analysis graph of it being shown in Fig. 3.

To perform the analysis, go to the menu bar *Analyse > Run*, or press **F9**. This simulates the working of the circuit, generates an analysis under a new tab, and produces a report of values and frequencies under *REPORTS* tab.

Click on the *Lock* button in the

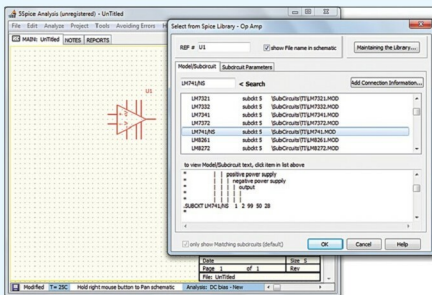


Fig. 4: Editing the parameters of op-amp

graph tab to prevent it from being overwritten by the next analysis. A multi-page schematic can be created in registered version.

The menu bar has a drop-down menu avoiding errors that cover almost all error possibilities.

One can also specify the model of a component so that 5Spice can simulate it properly. As an example, let us select an op-amp LM741. Right click

on it and select *Edit Parameters* from the drop-down menu. In the window that appears, the user can select LM741 from the given models of op-amps, as shown in Fig. 4. Then the manufacturer-specified parameters are loaded to the given op-amp. You can easily add other manufacturers' SPICE models to the library as 5Spice is compatible with most of them.

5Spice is a quick-and-easy tool for analogue circuit design and simulation at component level. The main window

Common Functions and Their Shortcuts

Function	Shortcuts
End or cancel current action	Right click or press Esc
Start wire	Ctrl+left click or press W
Junction	Double-click left or press J
Delete a component	Click on it and press Delete or press and hold Delete and select it
Edit general parameters of components	Press E
Edit values of components	Press V
Drag a component (this maintains connections)	Click and hold the left mouse button, move the mouse
Move a component (connections are not maintained)	Hold Shift and click left mouse button or click scroll wheel/middle button
Rotate a component	Press R
Flip/mirror vertically	Press F or M
Zoom in or out	Press I or O, respectively or hold Ctrl and rotate the mouse scroll wheel
Select or edit analysis	Press F8
Run the analysis	Press F9

is like a breadboard and the analysed graph acts as your oscilloscope screen, hence this is a very good way to test your circuits before building them on a breadboard. The complex analysis capabilities and simple user interface make it suitable for circuit designers at all levels—be it professionals or students. These qualities make 5Spice one of the best free tools available for circuit analysis. ●

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